

Arabic Legal Document Classification

CS476: Natural Language Processing | Course Project

Task A: Verdict Prediction

Task B: Legal Topic Classification

BONUS: Gradio Demo

Dataset: ALARB — 13,341 Saudi Commercial Court Cases (Hugging Face)

Instructor: Dr. Fatima AlShannaq

AI Disclosure: $\leq 30\%$ AI-assisted (prompts shared)

Why Arabic Legal NLP?

13,000+ Cases

Saudi commercial court records, newly digitized and available on Hugging Face as ALARB dataset.

Legal Arabic Challenges

Formal register, long documents (avg 181 words), Hijri dates, monetary figures, specialized statutes.

Real-World Impact

Supports judges in case routing, helps lawyers retrieve precedents, improves access to justice.

Underexplored Domain

Arabic legal NLP remains scarce. Most Arabic NLP benchmarks focus on news, MSA, or social media.

Dataset: ALARB

12,012

Training Cases

1,329

Test Cases

181

Avg Words / Case

4

Dataset Columns

Column Roles

Column	Task A	Task B
case_facts	Model Input	Model Input
court_reasoning	Analysis Only	Validation Only
applicable_laws	Optional	Label Source
verdict	Target Label	Not Used

Data Leakage Rule

court_reasoning contains the verdict rationale — using it as input would leak the answer.

applicable_laws references cited articles — using it as input for Task B leaks the topic label.

Data Preprocessing Pipeline

1

Character Normalization

Alef variants → ا, Taa Marbuta → ه, Hamza normalization

2

Diacritics Removal

Strip Tashkeel (short vowels) and Tatweel (.) using pyarabic

3

Legal Token Handling

Replace dates→[DATE], amounts→[AMOUNT], case IDs→[CASENUM]

4

Punctuation / Non-Arabic Removal

Remove all non-Arabic characters, punctuation, and digits via regex

5

Arabic Stopword Removal

NLTK list + domain stopwords: الدعوى, الدائرة, المحكمة

6

ISRIStemmer (NLTK)

Vocabulary: 37,920 → 9,671 tokens (74.5% reduction)

Feature Extraction Methods

Bag of Words (BoW)

CountVectorizer, max 10k features
Task A: 8,196 features
Task B: 8,666 features
Baseline; treats all terms equally

TF-IDF (uni + bigram)

Down-weights common legal boilerplate
Unigram: 8,196 features
Bigram: 10,000 features
Best overall feature set

N-grams

Word trigrams: 8,000 feat.
Char trigrams: 8,000 feat.
Char n-grams capture morphology
Word slightly better (Exp 3)

Word Embeddings (AraVec)

AraVec Twitter-CBOW 100d
Download failed → random fallback
OOV rate: 100% (domain mismatch)
F1-weighted: 0.4242 vs 0.6721 (TF-IDF)

AraVec (Twitter) has ~100% OOV on legal text — domain-specific embeddings (AraLegal-BERT) would be needed for meaningful embedding features.

Models Trained

Logistic Regression

Baseline

- L2 regularization, C=1.0
- solver='lbfgs', max_iter=1000
- TF-IDF unigram features
- One-vs-rest multi-class

LinearSVC (Best Model)

Best Model ★

- Grid Search: C ∈ {0.1, 1.0, 10}
- ngram_range ∈ {(1,1), (1,2)}
- 3-fold cross-validation
- Best: C=1.0, ngram=(1,2)

Feedforward ANN

Neural Model

- Dense(256, ReLU) → Drop(0.4)
- Dense(128, ReLU) → Drop(0.3)
- Softmax output layer
- Early stopping, patience=3

ANN architecture: 2,131,973 total parameters (8.13 MB). TensorFlow 2.19.0 on T4 GPU. Early stopping triggered at epoch 2 for both tasks.

Results — Task A: Verdict Prediction

Model	Accuracy	Prec (mac)	Recall (mac)	F1 (mac)	F1 (wtd) ★
LR + TF-IDF	0.6939	0.5925	0.4588	0.4999	0.6721
SVM + TF-IDF bigram ★	0.7157	0.5846	0.5184	0.5458	0.7072
ANN + TF-IDF	0.6912	0.5683	0.4821	0.5133	0.6802
LR + BoW	0.6624	-	-	0.5027	0.6594
LR + AraVec	0.5787	-	-	0.1466	0.4242

Best Model: LinearSVM

71.57% accuracy — TF-IDF bigrams + C=1.0 outperform all other configurations on Task A.

Class Imbalance Challenge

'Partially Accepted' (30 train examples) was never predicted correctly. Macro F1 (0.5458) is much lower than weighted F1 (0.7072).

Results — Task B: Legal Topic Classification

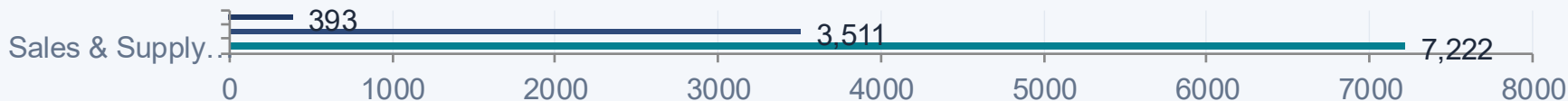
Model	Accuracy	Prec (mac)	Recall (mac)	F1 (mac)	F1 (wtd) ★
LR + TF-IDF	0.6986	0.6800	0.6029	0.6296	0.6800
SVM + TF-IDF bigram ★	0.6963	0.7132	0.6350	0.6674	0.6910
ANN + TF-IDF	0.6662	0.4302	0.3602	0.3245	0.5719

ANN Class Collapse on Task B

The ANN predicted almost all cases as 'Sales & Supply Contracts' (the majority class at 64.9%), achieving only F1-macro = 0.3245. This shows TF-IDF sparse vectors + dense neural networks are a poor fit without sequence information.

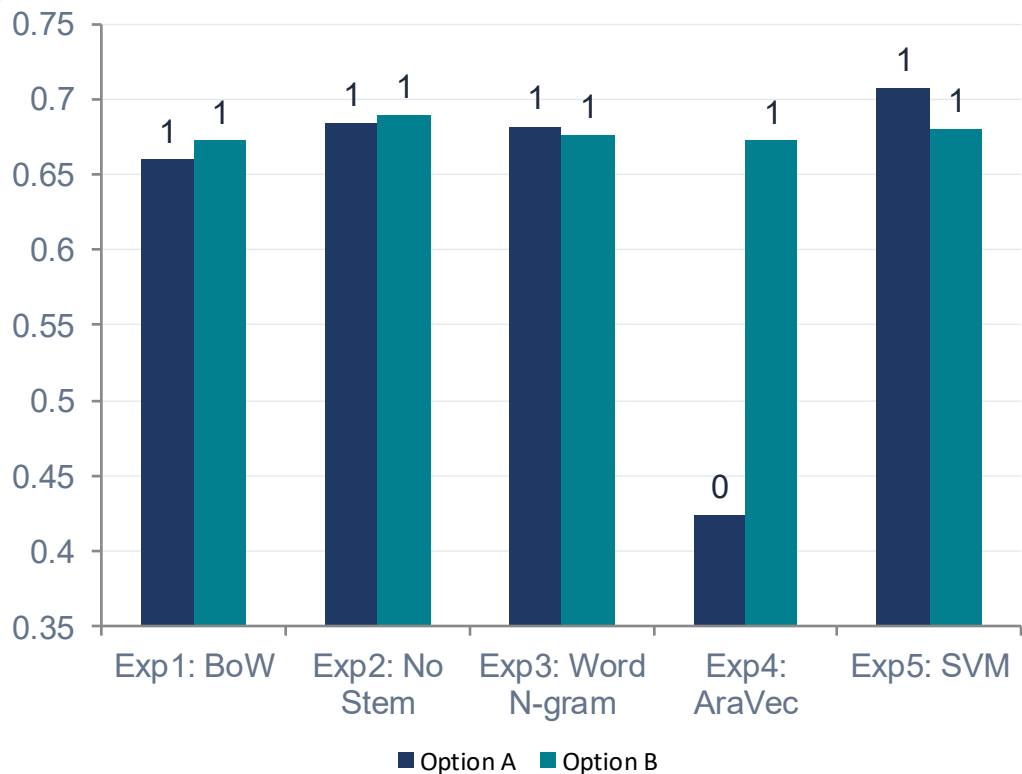
SVM Leads Task B

Highest precision (0.7132 macro) and F1-weighted (0.6910). SVM handles high-dimensional sparse TF-IDF well for both tasks.



Task B Label Distribution (imbalanced: 64.9% / 31.5% / 3.5%)

5 Experiment Scenarios



**Exp
1**

BoW: 0.6594 vs TF-IDF: 0.6721

→ TF-IDF +1.3%

**Exp
2**

No Stem: 0.6838 vs Stemming: 0.6895

→ Stemming +0.6%

**Exp
3**

Word N-gram: 0.6818 vs Char N-gram: 0.6761

→ Word +0.6%

**Exp
4**

AraVec: 0.4242 vs TF-IDF: 0.6721

→ TF-IDF +24.8%

**Exp
5**

LR: 0.6721 / ANN: 0.6802 vs SVM: 0.7072

→ SVM Best

Error Analysis

Task A (SVM Best) — Error Rate: 28.43% (533/1,875)

Error Pair	#	Cause
مقبولة كلياً → مرفوضة	233	Class imbalance
مرفوضة → مقبولة كلياً	143	Symmetric confusion
مقبولة كلياً → عدم اختصاص	48	Dismissal language similar
مقبولة كلياً → صلح	34	Settlement like acceptance

Task B (SVM Best) — Error Rate: 30.37% (676/2,226)

Error Pair	#	Cause
عقود بيع → نزاعات	363	Surface text overlap
نزاعات → عقود بيع	274	Symmetric confusion
عقود بيع → شراكات	21	Few training examples

Class Imbalance

Majority class (Fully Accepted, 57.8%) dominates predictions. 'Partially Accepted' (30 examples) is effectively unlearnable.

Language Overlap

Rejected and accepted cases share similar legal phrasing in case_facts. The discriminative signal is in court_reasoning (unavailable at inference).

Domain Boundary Blur

Task B categories are derived from cited statutes, not from surface text. Cases from different domains use almost identical language.

BONUS

Interactive Gradio Demo



Arabic Text Input

User pastes Arabic case facts in free text. No preprocessing needed from the user.



Task A Prediction

Returns predicted verdict class + confidence %
(LinearSVM calibrated via CalibratedClassifierCV).



Task B Prediction

Returns predicted legal topic + confidence %
simultaneously with Task A.



Public URL

Launched via Gradio share=True on Colab. Link:
<https://81f823306da0667705.gradio.live>

[OPTIONAL] AraBERT fine-tuning code is fully implemented (Section 16 of notebook). Set `RUN_BERT = True` with T4 GPU to activate.

Conclusion & Future Work

01

LinearSVM + TF-IDF bigrams is the best model for both tasks (Task A: 71.57% acc, Task B: 69.63% acc)

02

TF-IDF consistently beats BoW and random embeddings. Domain mismatch makes AraVec (Twitter) useless on legal text

03

Class imbalance is the #1 challenge. 'Partially Accepted' (30 examples) is unlearnable with standard methods

04

The ANN overfits on sparse TF-IDF vectors. A BiLSTM or AraBERT would better capture legal argument structure

Future Work: Fine-tune AraLegal-BERT | SMOTE for minority classes | BiLSTM for sequence modeling | Larger embedding vocabulary